Boundary-Aware Deployment for Cooperative Transport of Unknown Objects

Development manuscript — author information pending

***Abstract*— How far does deployment from local boundary observations carry a team toward actual object transport? We examine the complete perception–deployment–contact–transport–stop chain, separating geometric readiness, physical interaction, settling and spatial completion. A frozen development census compares boundary CVT, nearest-observed-target and local-spacing deployment under the same sensing, transport feedback, contact physics and evaluator. Nine shape–orientation conditions cover a 64-sided circle, a 24-vertex ellipse, a rectangle, an L-shape and a C-channel. Equal area, mass, robot initial states and target distance control several important confounders. With one fixed seed and no outcome-driven retries, the three methods complete 5/9, 2/9 and 5/9 tasks, respectively. A circle run passes the legacy G500 execution contract but misses the spatial goal by 1.177 m, demonstrating why projected arrival cannot replace spatial completion. Earlier feedback interventions and a new signed-error audit are retained as mechanism diagnostics. An anchored local readout rescues neither selected failure and worsens a successful C-shape control. The evidence characterizes capabilities and failure boundaries; it does not establish robust transport, statistical generalization, or a fully decentralized closed loop. The implementation combines local deployment with shared phase supervision.**

I. Introduction

A ring of robots around an unknown object is a preparation state, not a completed manipulation task. The team must perceive usable boundary information, distribute itself, generate actual contact forces, move along a requested displacement, and stop near a spatial goal. Failures can occur after apparently satisfactory coverage. Our research question is therefore: *under fixed sensing and transport control, how does boundary-based deployment affect executable transport across geometry and orientation?*

This question must be distinguished from novelty by demonstration. Cooperative transport and pushing span substantial prior work [18], [19]. Caging-based transport already addresses unknown shapes [2], and adaptive self-allocation has combined local detection, CVT and transport [1]. We do not claim that surrounding several shapes or using CVT is new. Instead, this development manuscript presents a controlled evaluation of an existing boundary-aware stack and identifies the evidence still needed for a defensible method contribution.

The principal comparison changes only the deployment target before the first CONTACT_READY event. Boundary CVT, nearest observed targets and local spacing otherwise share the same code and inputs. We freeze nine shape–orientation conditions and one seed, then retain all 27 started tasks, including watchdogs and adverse C-shape outcomes. The operational question is falsifiable: does boundary CVT produce more spatial completions than both existing deployment controls within this fixed census? Such a descriptive comparison can fail; it is not a population-level test.

The contribution at this stage is a reproducible system integration with explicit information and task contracts, accompanied by complete-task evidence across normalized shapes. The census tests the deployment-to-transport link and exposes its limits; it does not yet establish a superior new deployment algorithm. An earlier progress oracle rescued two selected failures, but a realizable anchored candidate did not. We close its expansion and retain the negative evidence and restricted mathematical diagnosis.

II. Related Work and Claim Boundaries

Caging restricts object escape through configuration-space geometry [6], [10]. Local-observation enclosing and grasping [3] and sequential caging [2] are particularly close to the present task. We use *boundary deployment* or operational enclosure for the measured arrangement. A surrounding image or high sampled coverage alone does not establish object closure.

Planar pushing [4], [9], adaptive collaborative control [5], [7], Force-ANTS [8], and grasp-wrench quality [20] connect coordination to manipulation under different assumptions. Sequential caging includes effective pushing and contact maintenance under initialization and communication assumptions. Song et al. combine detection-driven density, CVT and CBF coordination; their reported simulation setting differs from the present polygon/contact census. Tang et al. optimize contact modes and wrench feasibility using known object geometry and physical properties. These precedents preclude claiming novelty for local detection, CVT, or task-directed contact selection alone. Our two engineering deployment controls neither reproduce these approaches nor form a state-of-the-art benchmark.

CVT and coverage control provide the deployment background [14], [15]. Safety QPs relate to control-barrier methods [16], [17], but moving uncertain boundaries and sampled contact physics require assumptions beyond static-map results. Registration covariance and stable sampling [11], [12], [13] motivate the restricted information audit below. Neither a well-conditioned local matrix nor a correct numerical identity proves physically accurate cumulative progress.

III. System, Information and Task Contracts

A. Local deployment with shared supervision

Each robot observes nearby boundary samples and forms a local fused map using its own sensing and one-hop

observation exchange. The nominal deployment follows a boundary-weighted centroid in a local Voronoi region. In generic notation,

$$c_i = \frac{\int_{V_i} q\rho_i(q)\,dq}{\int_{V_i} \rho_i(q)\,dq}, \qquad u_i^{\text{deploy}} \propto c_i - p_i, \tag{1}$$

where the implemented density includes observed-boundary, directional-offset and exploration terms. The census preserves the existing density, discretization, gains and guards. The QP modifies the nominal command subject to the existing inter-robot and observed-object constraints; no thresholds are changed to improve outcomes.

The nearest control substitutes the closest observed offset target. The spacing control selects among locally eligible offset targets using neighbor separation and distance. Both preserve the CVT mass and deployment guards. After the first CONTACT_READY event, all three use the same subsequent controller. Earlier deployment can change contact configurations and downstream trajectories; shared code does not imply identical post-deployment states.

Local transport feedback uses the robot's own and neighboring usable progress estimates. A team-wide median progress and aggregate map coverage, readiness and speed signals drive a common monotone phase supervisor. The DBACT project name refers to decentralized boundary-aware cooperative transport as the research direction; this implementation does not yet establish a fully decentralized end-to-end loop. Shared task direction, neighbor communication and common phase state remain explicit assumptions. Object truth is available to simulated sensors, physics and independent evaluation, not to the local control readout. The earlier oracle is a separately labeled exception used only for diagnosis.

B. Five distinct events

CONTACT_READY uses a local-map distance criterion and a team quorum/dwell. In the frozen configuration, the distance band extends to 0.185 m whereas robot radius is 0.13 m. It therefore permits a geometric-ready state without physical contact. Physical contact requires penetration and contact force in the simulator. A separate recorded physical-readiness event requires sustained contacts with positive force projected on the task direction. It is stronger than the occurrence of one actual contact and is not assumed necessary for completion.

For target direction d, displacement L, initial object centroid x_0, and terminal centroid x_f,

$$J(t) = d^T(x(t) - x_0), \qquad e_{\text{end}} = \|x_f - (x_0 + Ld)\|\,. \tag{2}$$

Settling checks linear speed only, with no angular-speed criterion. It requires HOLD and speed at most 0.005 m/s for 40 consecutive control frames. Spatial completion requires settling *and* $e_{\text{end}} \le 0.12$ m. The unchanged control step is 0.05 s and watchdog is 3000 frames. A program finishing successfully is not a task completion.

G500 is a legacy execution contract, not a stricter version of spatial completion. Here it requires specified events and execution/safety checks, but arrival uses only $|J - L| \le 0.12$ m. Efficiency, direction error, cross-track deviation and yaw are diagnostic quality warnings, not pass/fail gates. Event deadlines are disabled, so its name does not impose a 500-frame limit. We report completion, G500 and trajectory quality separately. Discrete safety measurements and barrier-scaling counts are not continuous-time safety certificates.

IV. Frozen Full-Task Development Census

A. Geometry and fair comparison

The nine conditions are circle; ellipse long axis parallel/perpendicular to the task; rectangle long side parallel/perpendicular; L reference/rotated $180°$; and C opening toward/away from the goal. All world task directions are $+x$. The current CCTA rectangle, L and C outlines are retained as sources. The current C-channel opens in $+y$; rotations $-90°$ and $+90°$ produce the two conditions. L reference means its unrotated source polygon, not an ambiguous opening label. The frozen vertices and figure record each geometry.

Circle and ellipse reuse the historical deterministic 64- and 24-vertex outlines, rather than importing their old scenario settings. Every polygon is centroid-centered at $(4, 4)$ and isotropically normalized to area 2.05875 m^2. Areal density is 2 kg/m^2, giving mass 4.1175 kg. Actual polygon inertias are approximately 1.349, 1.550, 1.472, 2.087 and 2.547 kg m^2 for circle, ellipse, rectangle, L and C. Perimeter, reach, inertia and concavity remain different; equal area and mass do not equalize manipulation difficulty.

All conditions use 16 robots and the same legal scatter initial positions, generated once by seed 2 in a common annulus outside the maximum normalized reach. Actual initial-state and configuration hashes are verified for each three-method group. The goal distance is 1.2 m. Sensing, robot size, actuation, friction, contact stiffness, four physics substeps, control parameters and success rules are common. The team starts near the object, with some robots initially in sensing range: unknown full geometry is tested, not search from an entirely unknown remote location.

B. Budget, records and selection

The budget is exactly $9 \times 3 \times 1 = 27$ episodes. We do not replace seeds, retry failed tasks, extend watchdogs or tune parameters against outcomes. Source/environment snapshots, resolved configurations and product hashes identify this census separately from historical batches and v1/v2.

Every case records phases, contact, motion, braking, termination and safety. All 27 started tasks remain in the table. The nine conditions are within-block descriptive comparisons. The independent randomization unit is the complete shared scene/seed block, of which this census contains one. Frames and contact records are not independent trials; no reliability interval is reported.

Before execution, we select boundary CVT under the five reference orientations for the stage figure and overview video. All nine conditions have three-method videos synchronized by physical time and common view. Ended episodes retain their final measured frame with an explicit label.

TABLE I

ALL NINE CONDITIONS. ENTRIES: ENDPOINT ERROR IN METERS / SPATIAL COMPLETION (T OR F). ONE FIXED SEED PER CONDITION; THREE METHODS SHARE INITIAL STATES. FULL RECORDS ADDITIONALLY CONTAIN STAGE TIMES, SAFETY AND G500 REASONS.

Condition	CVT	Nearest	Spacing
Circle	1.177 / F	1.214 / F	0.906 / F
Ellipse parallel	0.838 / F	1.328 / F	0.903 / F
Ellipse perp.	0.356 / F	0.321 / F	0.394 / F
Rect. parallel	0.096 / T	0.217 / F	0.061 / T
Rect. perp.	0.039 / T	0.097 / T	0.030 / T
L reference	0.112 / T	0.134 / F	0.117 / T
L rotated	0.120 / T	0.306 / F	0.112 / T
C toward	0.101 / T	0.049 / T	0.046 / T
C away	0.846 / F	0.496 / F	0.818 / F

Videos use every fifth control frame at 10 fps (nominal $2.5\times$ playback), not episode-fraction synchronization. Estimated progress is labeled as the previous control update relative to the post-physics displayed pose; the new census mechanism figure instead aligns the estimate with the preceding, pre-physics object state. Raw laser rays are not retained: displayed teal samples are agent 00's saved fused map.

V. CAPABILITIES AND FAILURE BOUNDARIES

Table I reports every condition and method; Fig. 2 adds actual termination time and total robot path. Boundary CVT completes 5/9 tasks, nearest target 2/9, and local spacing 5/9. Corresponding G500 counts are 5/9, 1/9 and 3/9. These fixed-condition counts characterize this implementation; they are not success-rate estimates for arbitrary objects.

The proposed descriptive superiority criterion is not met: CVT ties local spacing in completion count. Circle and both ellipse orientations fail spatial completion under all three methods. Both rectangle orientations complete under CVT and spacing, while the nearest-target control completes only the perpendicular rectangle. L reference and rotated L complete under CVT and spacing; nearest-target fails in both. All methods complete with the C opening toward the goal, whereas all three C-away tasks exhaust the original 150 s watchdog and fail. The C-away endpoint errors are 0.846, 0.496 and 0.818 m for CVT, nearest and spacing. These results show a task-orientation boundary under this common configuration, not a claim that C shapes are intrinsically impossible.

In total, 22 episodes settle and five reach the watchdog: ellipse-parallel/spacing, rotated-L/nearest and all three C-away cases. Every episode reaches geometric CONTACT_READY and records physical contact and positive projected motion; none of those events suffices for completion. CVT and spacing achieve completion in the same five conditions but with different endpoint errors. These data support examining lateral motion and termination alongside deployment, rather than selecting a new feedback candidate solely from rank deficiency.

The circle/CVT example makes the contract distinction concrete. It reaches geometric readiness, produces actual contact and motion, and settles after 13.50 s. Its projected progress is 1.1247 m, within 0.12 m of the 1.2 m target. Its remaining hard execution gates also pass, so G500 passes. Yet lateral displacement yields $e_{\mathrm{end}} = 1.1772$ m. Recorded direction error is 46.25°, efficiency 0.6915, and maximum cross-track 1.1748 m. These create quality warnings, not G500 failures. Changing the contract to hide this discrepancy would destroy comparability; we retain both labels.

The supplementary five-shape event plate retains the preselected reference orientations and failures; all nine orientations have synchronized three-method videos. First physical contact is selected independently of the supervisor events. The figure sequence is event-indexed rather than an assumed chronological pipeline.

Figure 3 separates the actual team-progress readout, signed estimation error and true cross-track displacement. Census replay row $k+1$ stores the estimate computed before that step's physics; we therefore compare it with object truth at row k. All control samples are retained without smoothing. The circle approaches its projected goal while developing a large lateral offset; the C-away case instead stalls with a large longitudinal shortfall. These distinct trajectories do not establish a common causal estimation fault. Actual contact reports are from after robot motion but before cargo integration, and are labeled separately from control-time states.

VI. CLOSED FEEDBACK DIAGNOSTIC BRANCH

A. Oracle and negative local candidate

Nine archived rectangle/L/C development cases all reached geometric readiness and settled, but only five completed. None met the recorded positive-force quorum/dwell, despite completion in five cases. This rejects that particular recorded event as an unexamined necessary gate; it does not deny the need for physical interaction.

A six-episode diagnostic selected rectangle seed 2 and L seed 8 as failures and C seed 8 as a successful control. Replacing only local/team scalar progress with truth, from episode start, reduced endpoint errors from 0.1621 to 0.0720 m and 0.1516 to 0.0485 m in the failures. C error increased from 0.0083 to 0.0135 m. The oracle is not deployable, and post-hoc selection limits interpretation. Matching readiness times does not imply identical states at readiness; the intervention acts over the whole trajectory.

The subsequent v2 audit captured actual trimmed registration normals and fusion corrections. Of 22,812 registrations, 22,797 were task-direction identifiable at the frozen numerical tolerance. All 15 remaining records were already rejected by the original quality gate. Thus strict rank deficiency did not explain admitted measurements in these cases. Identifiability still does not imply accurate correspondences, negligible rotation or low accumulated bias.

A separate fixed nine-episode v2 comparison tested original feedback, a passive certificate and a local anchored readout. Original and certificate-only were identical in all 25 replay arrays and full step fields. The anchored readout used no object truth and retained the existing gain, map, velocity, downstream median and rejection rules. It rescued neither

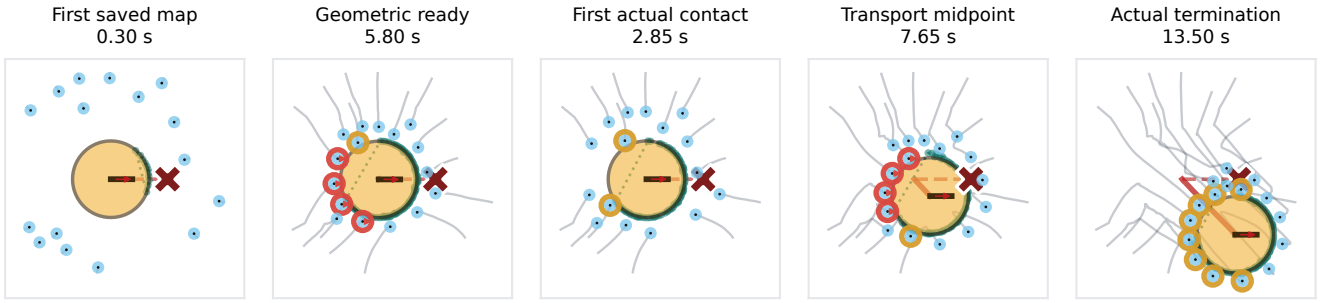


Fig. 1. Circle/CVT event-indexed process: first saved local map, geometric readiness, first physical contact, transport midpoint and actual termination. Event columns need not be chronological: contact here precedes readiness. Orange is evaluation geometry, teal is agent 00's fused map, and rings mark geometric readiness, not contact. The final task fails despite G500 passing.

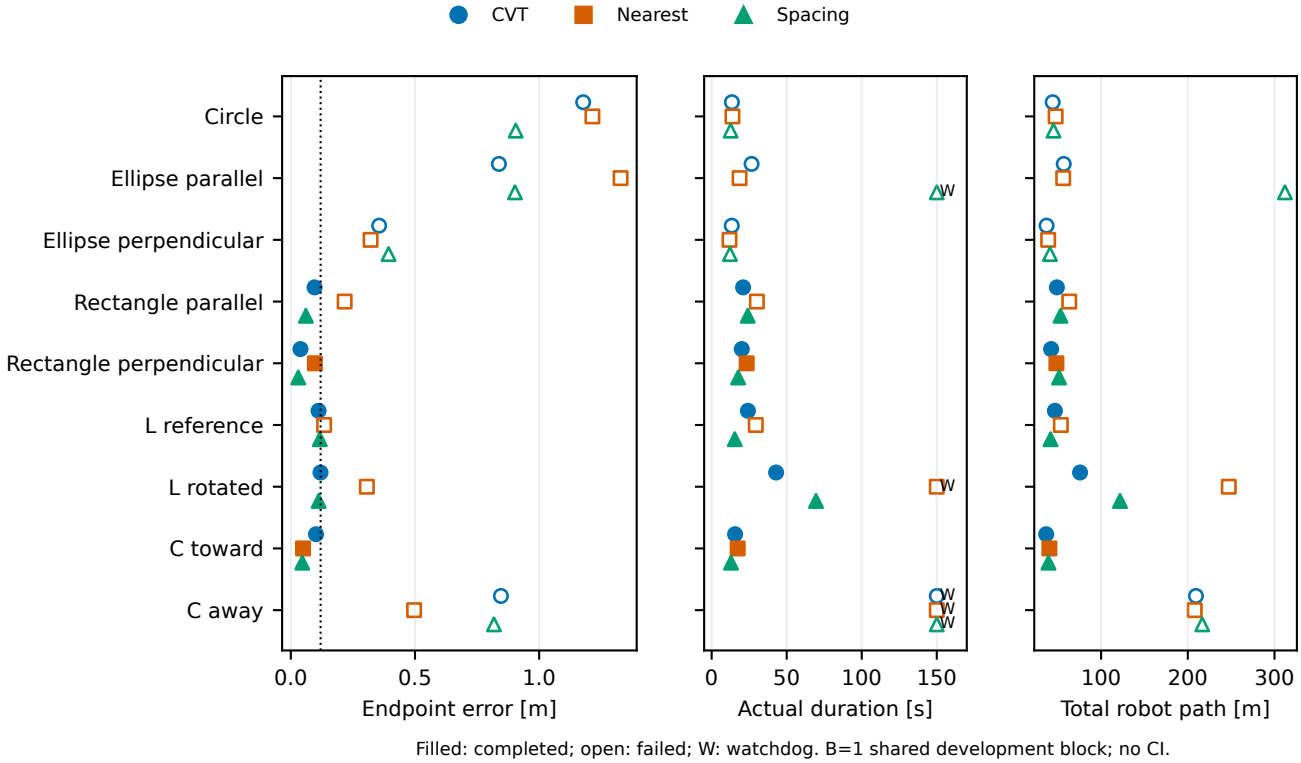


Fig. 2. All 27 tasks in one shared development block. Marker shape identifies method; filled/open markers denote spatial completion/failure. The dotted line marks the unchanged 0.12 m endpoint tolerance. Time is actual termination time, not a successful completion time for failed runs; W marks watchdog termination. Total robot path is computed from all recorded robot displacements. No confidence intervals or energy claims are implied.

failure. Rectangle endpoint error changed to 0.1592 m, L to 0.1522 m, and the successful C case worsened to 0.0187 m. Same-input local MAE was slightly worse in all three diagnostics. All nine G500 verdicts were false, including the spatially completed C cases. The candidate remains an experimental option, but expansion is stopped.

B. Restricted information analysis

For fixed point-to-plane correspondences and small pure translation,

$$b = A\Delta x + \epsilon, \qquad j = d^T \Delta x. \tag{3}$$

Proposition 1: The noiseless task scalar is identifiable for every displacement iff $d \in \mathrm{range}(A^T)$. When identifiable, minimum-norm exact weights are $\ell = A(A^T A)^\dagger d$, and $|\ell^T b - j| \le \|\ell\|\,\|\epsilon\|$.

Proof: Indistinguishable translations differ by $z \in \ker A$; their task scalars agree iff $d^T z = 0$ for every such z. Exact weights satisfy $A^T \ell = d$; the minimum-norm solution is as stated, and Cauchy–Schwarz gives the bound. □

This standard fact excludes uncertain normals, incorrect association, rotation and evolving maps. A circle's absolute rotation ambiguity does not imply that its translation is

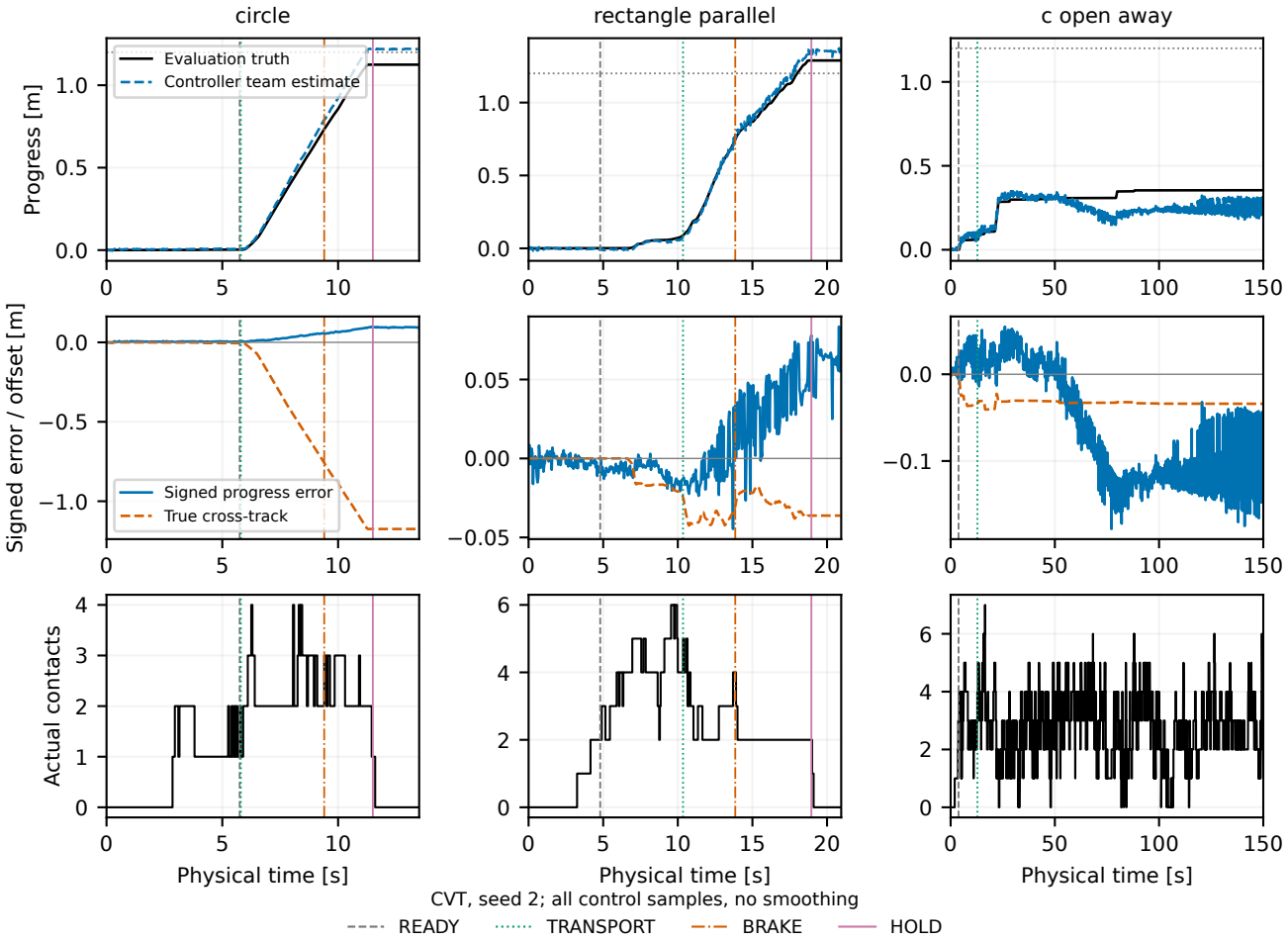


Fig. 3. Time-aligned CVT diagnostics for the circle, rectangle-parallel control and adverse C-away condition, selected after the census to illustrate distinct outcomes. Top: pre-physics evaluation truth and the actual shared team-progress estimate. Middle: signed progress error and true cross-track displacement. Bottom: actual contact reports. Event lines identify READY, TRANSPORT, BRAKE and HOLD; absent events are not imputed. Each column retains its actual time horizon, including the 150 s failure. One seed, no smoothing and no independent-sample inference.

unidentifiable. Polygon discretization is part of our executed model. The earlier synthetic observation-layout test is preserved as supplementary diagnosis; no new sensing policy is evaluated in the census.

The rejected candidate reads

$$\widehat{J}_k^a = g\big(d^T m_k^- + \ell_k^T b_k\big), \qquad g = 1.24. \tag{4}$$

It retains the historical map anchor. Under the restrictive rigid-map residual model $b_k = A_k[(x_k - x_0) - m_k^-] + \eta_k$, its error is $(g-1)J_k + g\ell_k^T\eta_k$. A correct certificate does not remove the gain term or bound real map/correspondence errors. This equation is a limitation, not evidence that gain alone caused the observed failures.

C. Signed cumulative accounting from existing logs

The offline audit uses the recorded pre-physics control-time truth. For robot i, define $u_{ik} = d^T m_{ik}^+$, held-readout term $h_{ik} = r_{ik}^{\text{original}} - g u_{ik}$, and candidate term $c_{ik} = r_{ik} - r_{ik}^{\text{original}}$. Then $r_{ik} = u_{ik} + (g-1)u_{ik} + h_{ik} + c_{ik}$. Median weights w_{ik} reproduce actual team progress T_k; tied values use an explicit agent-ID rule. The component allocation is not unique at ties and is not a causal attribution.

With $E_k = T_k - J_k$,

$$\begin{aligned}\Delta E_k = &\sum_i w_{ik}(\Delta u_{ik} - \Delta J_k) + \sum_i w_{ik}(g-1)\Delta u_{ik} \\ &+ \sum_i w_{ik}\Delta h_{ik} + \sum_i w_{ik}\Delta c_{ik} + \sum_i \Delta w_{ik} r_{i,k-1}.\end{aligned} \tag{5}$$

Accumulating all terms and the initial error reconstructs the signed error; omitting the voter-change term would misallocate it. Registration plus fusion also reconstructs map increments to numerical precision. Both are accounting checks, not physical-estimation proofs.

The residual of cumulative reconstruction is below 10^{-12} m in every arm. The different signed terms can compensate or reinforce each other; they do not isolate an implementation defect. The raw-map, gain, hold and endogenous voter effects remain hypotheses for future targeted tests, not a reason to tune this completed batch.

TABLE II

BASELINE CUMULATIVE SIGNED-ERROR ACCOUNTING, MILLIMETERS, AT THE LAST ALIGNED CONTROL INSTANT. COMPONENTS USE A FIXED MEDIAN TIE RULE AND ARE NOT CAUSAL EFFECTS. DISPLAYED TERMS ARE ROUNDED.

Case	Map–truth	Gain	Held	Voters	Total
R2	-18.9	306.0	-13.9	-102.4	170.8
L8	-168.7	236.4	36.6	50.7	154.9
C8	-240.0	255.7	22.5	-2.1	36.0

For the v2 baseline episodes, the first shared HOLD in rectangle and L occurs with true speeds zero but endpoint errors outside the original task tolerance. Estimated/true progress is $1.4651/1.2939$ m against $L = 1.4467$ m for rectangle, and $1.3082/1.1535$ m against $L = 1.3044$ m for L. C has already reached the projected goal. These observations identify premature spatial stopping in the selected failures. Individual transport-loop hysteretic holding states were not saved; their exact first transitions cannot be reconstructed from scalar input alone. No new gain, certificate or candidate is introduced.

VII. LIMITATIONS AND RESEARCH DECISION

This near-field census uses one robot arrangement and common $+x$ goals, not independent generalization data. Equal area and mass leave inertia, perimeter and contact opportunities unequal. Penalty contact, 0.01 m range noise, shared supervision and simulated localization restrict external validity. No hardware validation is claimed.

Changed deployment also changes trajectories and contacts; it cannot assign all failures to feedback. Oracle/v2 cases cannot be pooled with this differently normalized census. Algebraic information and bookkeeping results establish neither transport stability nor continuous safety.

The progress branch is a mechanism diagnostic and limitation, not the main method contribution. A stronger contribution still requires a deployable mechanism with matched information/action budgets, evidence against the closest work, and independent validation. Noise, robot-resource and new-seed designs have not been executed and are not results of this paper. Future candidates should target a supported mechanism with a predeclared falsifier. The current anchored candidate is closed, including expansion across shapes.

VIII. CONCLUSION

The fixed 27-task census connects unknown-boundary deployment to actual contact, transport and stopping. Smooth and nonconvex objects expose distinct failure boundaries, while legacy execution checks remain separate from spatial completion. Negative feedback and full-task results constrain the next mechanism hypothesis.

REFERENCES

[1] J. Song, Y. Bai, and N. Wakamiya, "Cooperative transportation without prior object knowledge via adaptive self-allocation and coordination," arXiv:2602.19070, 2026.

[2] V. S. Vardharajan, K. Soma, and G. Beltrame, "Collective transport via sequential caging," arXiv:2106.03132, 2021.

[3] Y. Kobayashi and S. Hosoe, "Cooperative enclosing and grasping of an object by decentralized mobile robots using local observation," *Int. J. Social Robotics*, vol. 4, pp. 19–32, 2012, doi: 10.1007/s12369-011-0118-7.

[4] Z. Tang, Y. Feng, and M. Guo, "Collaborative planar pushing of polytopic objects with multiple robots in complex scenes," in *Robotics: Science and Systems XX*, 2024, doi: 10.15607/RSS.2024.XX.063.

[5] L. Yan, T. Stouraitis, and S. Vijayakumar, "Decentralized ability-aware adaptive control for multi-robot collaborative manipulation," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 2311–2318, 2021, doi: 10.1109/LRA.2021.3060379.

[6] G. A. S. Pereira, M. F. M. Campos, and V. Kumar, "Decentralized algorithms for multi-robot manipulation via caging," *Int. J. Robotics Research*, vol. 23, nos. 7–8, pp. 783–795, 2004, doi: 10.1177/0278364904045477.

[7] P. Culbertson, J.-J. E. Slotine, and M. Schwager, "Decentralized adaptive control for collaborative manipulation of rigid bodies," *IEEE Trans. Robotics*, vol. 37, no. 6, pp. 1906–1920, 2021, doi: 10.1109/TRO.2021.3072021.

[8] Z. Wang and M. Schwager, "Force-amplifying N-robot transport system (Force-ANTS) for cooperative planar manipulation without communication," *Int. J. Robotics Research*, vol. 35, no. 13, pp. 1564–1586, 2016, doi: 10.1177/0278364916667473.

[9] C.-Y. Chai, W.-H. Peng, and S.-L. Tsao, "Object rearrangement through planar pushing: A theoretical analysis and validation," *IEEE Trans. Robotics*, vol. 38, no. 5, pp. 2703–2719, 2022, doi: 10.1109/TRO.2022.3153785.

[10] G. Wang, K. Ren, A. S. Morgan, and K. Hang, "Caging in time: A framework for robust object manipulation under uncertainties and limited robot perception," *Int. J. Robotics Research*, vol. 44, no. 14, pp. 2365–2389, 2025, doi: 10.1177/02783649251343926.

[11] A. Censi, "An accurate closed-form estimate of ICP's covariance," in *IEEE Int. Conf. Robotics and Automation*, pp. 3167–3172, 2007, doi: 10.1109/ROBOT.2007.363961.

[12] S. Bonnabel, M. Barczyk, and F. Goulette, "On the covariance of ICP-based scan-matching techniques," in *American Control Conference*, pp. 5498–5503, 2016, doi: 10.1109/ACC.2016.7526532.

[13] N. Gelfand, L. Ikemoto, S. Rusinkiewicz, and M. Levoy, "Geometrically stable sampling for the ICP algorithm," in *Fourth Int. Conf. 3-D Digital Imaging and Modeling*, 2003.

[14] Q. Du, V. Faber, and M. Gunzburger, "Centroidal Voronoi tessellations: Applications and algorithms," *SIAM Review*, vol. 41, no. 4, pp. 637–676, 1999, doi: 10.1137/S0036144599352836.

[15] J. Cortés, S. Martínez, T. Karatas, and F. Bullo, "Coverage control for mobile sensing networks," *IEEE Trans. Robotics and Automation*, vol. 20, no. 2, pp. 243–255, 2004.

[16] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs for safety critical systems," *IEEE Trans. Automatic Control*, vol. 62, no. 8, pp. 3861–3876, 2017, doi: 10.1109/TAC.2016.2638961.

[17] L. Wang, A. D. Ames, and M. Egerstedt, "Safety barrier certificates for collisions-free multirobot systems," *IEEE Trans. Robotics*, vol. 33, no. 3, pp. 661–674, 2017, doi: 10.1109/TRO.2017.2659727.

[18] E. Tuci, M. H. M. Alkilabi, and O. Akanyeti, "Cooperative object transport in multi-robot systems: A review of the state-of-the-art," *Frontiers in Robotics and AI*, vol. 5, Art. 59, 2018, doi: 10.3389/frobt.2018.00059.

[19] J. Stüber, C. Zito, and R. Stolkin, "Let's push things forward: A survey on robot pushing," *Frontiers in Robotics and AI*, vol. 7, Art. 8, 2020, doi: 10.3389/frobt.2020.00008.

[20] C. Ferrari and J. Canny, "Planning optimal grasps," in *IEEE Int. Conf. Robotics and Automation*, pp. 2290–2295, 1992.